

Vision systems research

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Abstract—Aperture Open is a team of competitors of the RoboCup Junior Soccer Vision (formerly Soccer Open) category, representing Colégio Objetivo from São Paulo city, Brazil. This paper explains the methods available to achieve a 360° vision, presents tests made by the team, evaluates advantages and disadvantages and explains the new configuration with two cameras that the team developed.

Key-words—RoboCup, Soccer Vision, Mirrors, Cameras, Lens.

I. INTRODUCTION

In the Soccer Vision category of RoboCup Junior, it is essential for a robot to be able to locate the ball on the field by effectively viewing it with a camera. For this, different methods were developed to achieve a 360° vision so the robot always sees the ball.

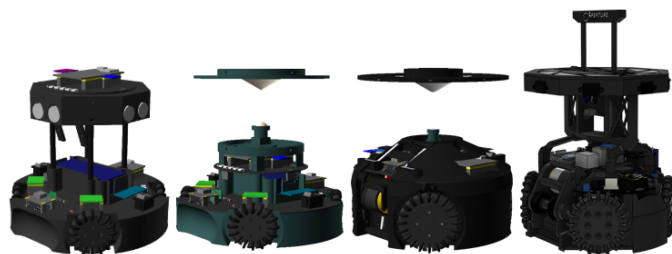


Fig. 1. Different robots made by our team, going from a front camera, to a camera with mirror, to two cameras with fisheye lenses.

To help new Soccer Vision teams to implement their own vision system and considering the ball change of Soccer Infrared, that encouraged Infrared teams to develop vision systems to detect the new orange ball, we created this document gathering tests made with different materials available, explaining the production process, advantages and disadvantages, and also explaining the vision system we developed to balance cost, processing usage and production process.

II. MIRRORS

A. Overview

The most common method used to achieve a 360° vision is by using a camera pointed to a conical mirror. This way it is possible to get a circular image of the entire field using only one camera. The main ideas to produce a conical mirror are divided into two types: molding and polishing a metal part, and heating and molding a mirrored plastic sheet. We also tested alternative methods, such as using mirror sprays.

B. Metal mirrors

The central idea is to get a evenly curved metal part and polishing it until it achieves a satisfactory reflection. The common method is buying a metal cylinder, molding it on a lathe and then sanding and using polishing compound. The cylinder can be either of aluminum, a cheaper and lighter material, or of stainless steel, that is heavier and more expensive, but achieves a more consistent reflection and preserves it better. After buying the cylinder, the team needs to mold it using a lathe to achieve a shape that fits their own robot, considering camera and mirror position planed on the robot, so the camera can view from the area next to the robot, to the walls of the field. Then, it is necessary to use sandpapers to remove imperfections, use polishing paste to give the final polishing and make it actually reflect. The production process can also be ordered with CNC companies, such as JLCCNC, although it is more expensive.

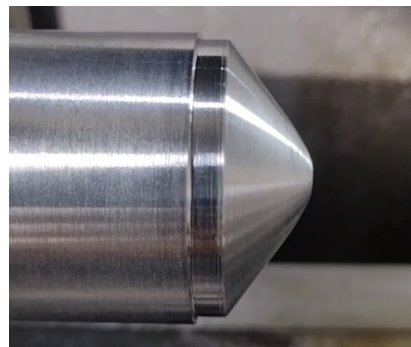


Fig. 2. Aluminum cylinder molded on a lathe.

The cheapest and most accessible option for metal mirror we found is buying a hemisphere, half of a sphere, mold of aluminum. Being a spherical shape, it does not require custom production, just height adjustment to fit the camera's FOV. The rest of the production is the same as the conical mirror, with sanding and polishing process. The problem with this material is that it does not achieve a reflection as good as using the cylinder.



Fig. 3. Polished aluminum mold.

C. Plastic sheet mirrors

By heating a plastic sheet that has a mirrored coating, it is possible to mold it by pressing the sheet against a 3D printed conical mirror. This method results in a more distorted but better quality image.

The main problem is the material of the mirrored sheet used. Normally, the material used is a PVC (Polyvinyl Chloride) or PS (Polystyrene) sheet mirrored using by vacuum a thin layer of aluminum. Since this type of material was not available in Brazil, we tried using a PVC plate coated with acrylic and mirrored with a cheap material, but although we managed to heat and mold it in the conical shape, the material did not retain its reflexive properties after the process.



Fig. 4. PVC plate of cheap coating after heating.

The closest we got to a conical mirror using cheap accessible mirrored sheets was using an imported mirrored sheet made of PET (Polyethylene Terephthalate), that we cut in a circle and fold it on an 3D printed cone, as the PET did not deform properly after heated. This could serve as an alternative method for countries where the PVC or PS sheets mirrored with aluminum are not available.

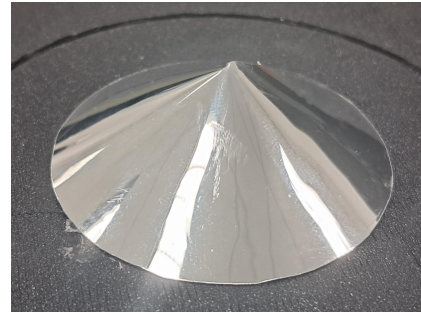


Fig. 5. Folded PET mirrored sheet.

D. Other methods tested

Another tests include the use of a chrome spray, spray mirror and mirrored insulfilm, all with insufficient outcomes.

One of the ideas we tested was creating a mirrored pyramid, but not only it had too many blind spots, depending on the material used, it is difficult to produce. Still, an adapted version of this method could be availed for Infrared teams until last year. By folding a square of the mirrored sheet in half, it is possible to divide the image of the camera, possibly allowing the camera to view both goals with minimal manufacturing necessary. As now many teams are considering using conical mirrors to view the ball, this method is indeed not ideal.

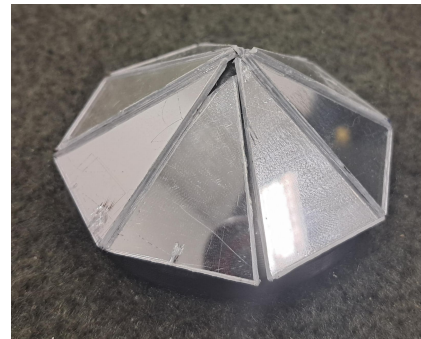


Fig. 6. Pyramid of PVC mirrored plates.

III. ALTERNATIVE METHOD PROPOSED

The previous section explained different methods to achieve a 360° vision using mirrors, and a good part about them is that various teams from around the world can serve as examples of what to be followed to produce a mirror. But the main problem is that mirrors require a manufacturing process, that as it can be noticed with the different tests we made, either has high cost or depends on material availability to achieve ideal results.

Considering that, we developed a new vision system that uses two cameras with omnidirectional/fisheye lenses to achieve a 360° vision while also making changes to increase the space available on the robot and save processing usage. Allow us to explain.

Until 2022, an omnidirectional lens, defined as having a FOV higher than 140° horizontally and 80° vertically, was only allowed if produced by the own students, and only one

camera per robot was allowed. Until today most of the teams kept using conical mirrors to achieve a 360° with only one camera of low FOV, but since 2023, the rules were changed to "A robot may use any number of cameras without restrictions on lenses, optical parts, optical systems, and total field of view. Components may be sourced in any way the team sees fit."¹.

An example of method that was created after this rule is the use of four cameras with $90\text{-}100^\circ$ horizontal FOV. The idea is that the FOVs are combined to achieve a 360° vision. This method is interesting because not only it removes the necessity of a manufacturing process, it also allows components and supports to pass in the space between the cameras, that previously would need to be free so the camera vision of the mirror would not be harmed. On the other hand, four cameras is also an expensive option, and it also means four times the processing power required.

So, considering the advantages and disadvantages of the different methods, to achieve a 360° vision our robot has two cameras, one pointing to robot's forward (ball capturing zone) and the other backwards. As it is possible to view in the image below, both cameras FOVs allow the 360° vision creating blindspots only in the area of the robot's structure, a non required vision as the camera only needs to view the field.

The two cameras are of sensor imx307, of 2MP and a resolution of $1920 \times 1080p$, with a max frame rate of 60Hz. It also has customisable saturation, white balance, gain, contrast, exposure time, brightness and frame size, which is fixed on a non-volatile memory.



Fig. 7. imx307 camera sensor with module.

The lens used is a fisheye of 1.4mm. In the camera used, considering the sensor and image sizes, it was measured that the lens allows 210° FOV horizontally and 150° vertically.

Distancing the cameras to maximize this blind spot on purpose, it is possible to create an area that allows passing wires and structural parts without creating a single blind spot on the cameras view of the field. Finally, instead of tilting down the cameras, which would increase the image area used to view the field and allow the cameras to be distanced further without creating blind spots, the cameras vision is aligned parallel to the field to maximize the horizontal FOV of each camera, essential to allow the use of only one camera at a time, reducing the processing usage.

¹RoboCupJunior Soccer Rules; 2023; p. 4; in: <https://tinyurl.com/msw9hnae>;

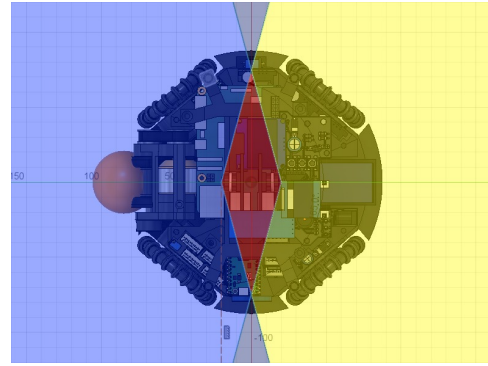


Fig. 8. Top view of FOVs. Blue: front camera view; Yellow: back camera view; Red: blindspots; Gray: cameras intersection.

To better explain the processing usage reducing, a common configuration used in Soccer Vision is having two cameras, one pointed to the mirror and one pointed forward. With this setup, it is not possible to use only one camera at a time, as they have different purposes. The other method, using four cameras, limits the possibilities of processing usage reducing, as the cameras FOVs are smaller.

Finally, as the cameras of our robots are located in the top of the robot, it is possible to view the field in its entirety, including the areas close to the robot. With this positioning the distortion is minimized when compared to using a mirror, as the cameras get a rectangle view of the vision pointing forward/backward, while with the mirror it would be a distorted circular image.

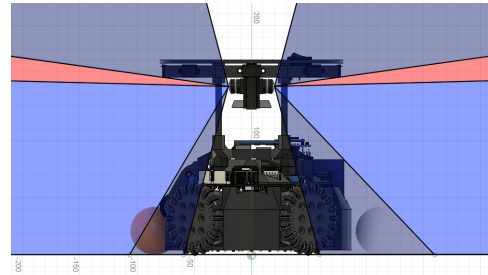


Fig. 9. Side view of FOVs. Gray: vision blocked by the robot's structure; Red: areas cut in code; Blue: usable view of the field.

IV. CONCLUSION

So, after analyzing different methods used to achieve a 360° vision, the method our team proposes balances the advantages and disadvantages of either using a mirror or using four cameras to achieve a more cost benefit option that improves space usage, processing power and image quality.

This is a method that few teams had properly tested, but that we have developed and determined it could replace our mirrors. At the same time, different tests we made did not produce sufficient outcomes. What we want the teams that are reading this document to consider, after explaining both the conventional methods and the alternative methods we tested is that, although some techniques are now consolidated and used

by most of the teams, it is also possible to innovate, search and develop ideas that could even surpass the consolidated ones.

V. COMPLEMENTARY LINKS

GitHub repository for our documentation:
<https://github.com/Maizeninha/AptureOpen2026-Docs>

VI. BIBLIOGRAPHY

METHODS RESEARCH SOURCES

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